



V93000 Specifications

Pin Scale and Smart Scale Infrastructure

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Infrastructure specifications - V93000 Pin Scale and Smart Scale systems

Scope of specifications

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board, *valid system calibration* and a fixture delay measurement (TDR), these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.

Test programming and debug

The SmarTest production test environment includes manufacturing test interfaces to operators, handling equipment and factory networks. A graphical testflow editor links device tests into a production-ready test program, where the tests are set up via fill-in-the-blank test functions. User-specific tests are programmed with test methods in C++. Links are available for design-to-test conversion. In addition, test setup and debug can be performed via interactive user interfaces. Result analysis tools are available for error locations, timing behavior, analog waveforms, shmoo plots, pin margin tests and memory bit map.

Supported channel cards

- Different digital cards can be mixed with DC cards within one card cage of the test head.

Compatibility

- Software is compatible between all test heads.
- The same DUT boards fit on all test heads. The outer pin groups of a large DUT board are not supported on the small DUT interface.
- The docking system is identical for all test heads. However, some docking pin positions differ between the large and the small/compact test head. The A-test head has no hard docking.

Operating system

Linux Red Hat Enterprise is used in all test head configurations. This applies to all test heads with small DUT interface and DUT Scale interface.

Calibration and operation

Warm-up time	60 minutes
Basic maintenance period	6 months
Base calibration period (traceable calibration)	6 months
Calibration period (system auto adjustment)	3 months ^[1]
DC update period	2 weeks
DC update (SmarTest 7) / Enhanced Accuracy Adjustment (SmarTest 8) period for enhanced accuracy of AVI64	1 week

^[1] Valid at ambient temperature within ± 2.5 K of calibration temperature.

Further information

For more information about the Advantest V93000 SOC Series, please visit www.advantest.com.

Contact information

For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

Test heads

Maximum channel count	A-Test Head	Compact Test Head ^[2]	Small Test Head ^[2]	Large Test Head
Number of card cages	1	2	4	8
Pin Scale 1600	1024	2048	4096 ^[3]	4096

The use of other digital channel cards, DC cards, analog instruments or RF cards may reduce the maximum channel count.

^[2] Applies if the DUT Scale interface is used.

^[3] Divide by 2, if small DUT interface is used

Device power supplies (channel slots)

Supported channel slot device power supplies

Maximum channel count	A-Test Head ^[4]	Compact Test Head ^[4]	Small Test Head ^[4]	Large Test Head
Number of card cages	1	2	4	8
DC Scale DPS128-HC (E8023CS)	1024	2048	2048	2048
DC Scale DPS64-HC (E8023CSH)	512	1024	2048	2048
DC Scale UHC4T (E8024UH)	32	64	128	256

Note:

- Mixing different device power supply types within one Test Head is supported.
- Refer to *DC Scale device power supply* specification sheet for more information.

^[4] Applies to Test Heads with the small DUT interface and with the DUT Scale interface.

Parametric measurements

Voltage specifications at utility pogo block

Range	Voltage range	Resolution	Accuracy of reading/ setting
Voltage range (force/ measure)	-5 V to 7.5 V ^[5]	250 μ V	± 2 mV ^[6]
Maximum input voltage range	-6 V to 9 V		

Current specifications at utility pogo block

Range	Current range	Resolution	Accuracy of reading/ setting
Range 1	± 200 mA	6 μ A	± 200 μ A $\pm 0.1\%$
Range 2	± 5 mA	250 nA	± 10 μ A $\pm 0.1\%$
Range 3	± 200 μ A	6 nA	± 200 nA $\pm 0.1\%$
Range 4	± 5 μ A	250 pA	± 10 nA $\pm 0.1\%$

Direct DVM Access (at utility pogo block)

- Supported DMM: Keysight 3458A with option 002
- Absolute voltage measurement accuracy: $\pm 40 \mu\text{V} + \text{DMM specification}$
- Input voltage range: -5 V to 7.5 V

^[5] Over-programming up to +8 V.

^[6] Add $\pm(I_a * R)$ if connected through digital pin. I_a is the actual current; R is the wiring resistance of the digital channel. See specification of digital card for potential additional card-specific terms.

Utility resources

Utility lines of V93000 Pin Scale and Smart Scale systems allow you to control external devices such as relays, solenoids or indicators, depending on the application. The lines are implemented as FET switch outputs.

The utility power supplies of V93000 Pin Scale and Smart Scale systems support DUT boards with additional active components. Typical examples for these active components are relays, switches, operational amplifiers, and buffers.

Utility lines (V93000 systems)

Utility lines allow you to control external devices such as relays, solenoids or indicators, depending on the application.

The utility line configuration and characteristics of V93000 Pin Scale and Smart Scale systems is defined in this section. For the utility line configuration and characteristics of V93000 EXA Scale systems, refer to [#unique_12](#).

Utility lines 129-256 are optional and supported from SmarTest release 7.2.0 on.

Characteristics		A-test head, Compact test head, Small test head		Large test head
		Small DUT IF	DUT Scale IF	
Configuration				
Number of utility lines		64	256	256
Pogo block		Utility 1 ... 4	Utility 1 ... 8	Utility 1 ... 8
Output				
Type		Open collector		
Maximum current		60 mA (each)		
Voltage	Open collector output open, no load applied	The maximum voltage and the impedance depend on the configuration of the system: • An internal pull up resistor is available per utility line. • The pull-up resistor is connected to the internal supply voltage of the utility line. • The resulting output voltage of the open output depends on the configuration of the system and will be in the range from 3.3 V to 5.0 V. We recommend using external pull up resistors for applications with higher requirements.		
	Open collector output closed	Typically ~0.7 V at low collector currents (<10 mA) or up to 1.0 V at higher collector currents		
Input				
Type		TTL		
Voltage range		0.0 V ... 5.0 V		
Impedance		> 10 kΩ		

Utility power supplies (V93000 systems)

The utility power supplies support DUT boards with additional active components. Typical examples for these active components are relays, switches, operational amplifiers, and buffers.

Internal utility power supply

		A-test head		Compact test head		Small test head		Large test head
		small DUT IF	DUT Scale IF	small DUT IF	DUT Scale IF	small DUT IF	DUT Scale IF	
Configuration								
Number of card cages	1	1	2	2	4	4	8	
Total number of supplies	1	1	1	2	1	4	5	
3.2 A supply routed to								
1 utility pogo block	-	-	-	-	-	2	4	
2 utility pogo blocks	-	-	-	-	-	1	-	
4 utility pogo blocks	1	-	1	2	1	1	1	
8 utility pogo blocks	-	1	-	-	-	-	-	
Characteristics								
Output voltage	4.85 V +1%, -5% if slots one or two in the card cage supplying the utility power are not empty and not populated with PS3600, PS-HX or PS9G 4.85 V +1%, -20% if slots one and two in the card cage supplying the utility power are empty							
Total output current (max. 1 A per block)	3.2 A	3.2 A	3.2 A	6.4 A	3.2 A	7.2 A	7.2 A	
Pogo pins	+4.85 V: UP5V GND: UGND							

External utility power supply (E7002EUSR/E7002EUSM)

The External utility power supply is optional and supported from SmarTest release 7.2.0 on.

Characteristic	DUT Scale IF (or the large DUT IF) on all types of test heads	
Output voltage	5.0 V, sensed, -0 %, +3 % (+5.15 V) 3.3 V, sensed, -0 %, +3 % (+3.4 V) -5.0 V, sensed, -0 %, +3 % (-5.15 V)	
Output current (max. 2 A per block and for each voltage)	5.0 V:	8 A
	3.3 V:	6 A
	-5.0 V:	6 A
Pogo pins	The optional external utility power supply is available only on the 4-row utility pogo blocks.	
	5.0 V:	XPP5V and XPP5Vs
	3.3 V:	XPP3V3 and XPP3V3s
	-5.0 V:	XPM5V and XPM5Vs
	GND:	UGND and UGNDs

Characteristic	DUT Scale IF (or the large DUT IF) on all types of test heads	
	Use only one sense line per voltage and only one ground sense line for power sensing. Leave the sense lines on the other utility blocks unconnected. If the power line is not used, leave all sense lines unconnected. This restriction applies to XPP5Vs, XPP3V3s, XPM5Vs, and UGNDs.	
Power OK output	XDCGD	

Internal Utility Power Supply (SE-UDPS) (E7002US)

The optional "Internal Utility Power Supply (SE-UDPS)" is a product from Salland Engineering (Europe) BV. The SE-UDPS is routed to a 3-row DPS-slot pogo block.

	A-test head		Compact test head		Small test head		Large test head
	small DUT IF	DUT Scale IF	small DUT IF	DUT Scale IF	small DUT IF	DUT Scale IF	
Maximum number of supplies	1	1	2	2	4	4	8
Configuration	- Any DPS slot can be used - The Internal Utility Power Supply can be mixed with any other device power supply in a test head, no limitations apply. - Diagnostics requires the diagnostics board E8026DIAG.						
Characteristics							
Output voltage and current	Voltage	Accuracy		Maximum current			
	+5.0 V	±2 %		3 A			
	+12.0 V	±2 %		1 A			
	+15.0 V	±2 %		1 A			
	-15.0 V	±2 %		850 mA			
Pogo pins	The SE-UDPS is routed to a 3-row DPS-slot pogo block.						
	+5.0 V:	P1, PS1, G1 (two pins)					
	+12.0 V:	P2, PS2, G2 (two pins)					
	+15.0 V:	P3, PS3, G3 (two pins)					
	-15.0 V:	P4, PS4, G4 (two pins)					
General capabilities	- Overcurrent protection - Individual ON / OFF per power supply - ON / OFF detection when DUT board is removed - Kelvin connection per DPS channel (power line)						
Power on settling time	Condition: Output voltage within ±5 %-band of nominal voltage, 47 μF external capacitive load						
	+5.0 V:	2 ms					
	+12.0 V:	6 ms					
	+15.0 V:	3 ms					
	-15.0 V:	3 ms					

	A-test head		Compact test head		Small test head		Large test head
	small DUT IF	DUT Scale IF	small DUT IF	DUT Scale IF	small DUT IF	DUT Scale IF	

Load capacitance range	0 μ F to 470 μ F per channel						
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Miscellaneous characteristics

Utility lines and utility power supplies

For more information, refer to *Utility lines (V93000 systems)* on page 8 and *Utility power supplies (V93000 systems)* on page 9.

DUT board disconnect lines

Characteristic	A-test head		Compact test head		Small test head		Large test head
	small DUT IF	DUT Scale IF	small DUT IF	DUT Scale IF	small DUT IF	DUT Scale IF	
Number of card cages	1	1	2	2	4	4	8
Number of disconnect lines	4	8	2	8	4	8	8

Side panel port

	A-test head	Compact test head ^[7]	Small test head ^[7]	Large test head
Number of card cages	1	2	4	8
DC rails	1	2	4	8

Clock domains

	A-test head	Compact test head ^[7]	Small test head ^[7]	Large test head
Number of card cages	1	2	4	8
Number of clock domains	1	2	4	8

Reference clock

Reference clock output	
Frequency	10 MHz
Impedance	50 Ω
Output connector	BNC
Output signal	0 V to +2 V, into open

Reference clock (10MHz) value accuracy

- Digital systems: < 15ppm over lifetime, no external adjustment (digital base calibration) required

^[7] Applies to test heads with the small DUT interface and with the DUT Scale interface.

Reference clock input	
Frequency	10 MHz
Impedance	50 Ω
Input connector	BNC
Input signal	RF 0 dBm - 10 dBm, into 50 Ohm

External instruments (compact test head)

External input available to connect three external instruments.

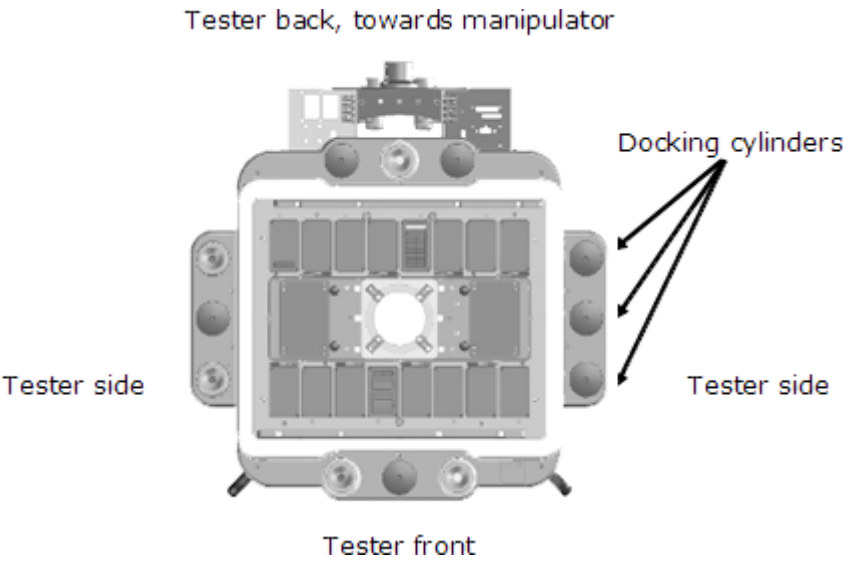
DUT interface types

Interface type	A-test head	Compact test head	Small test head	Large test head
Small DUT interface	x	x	x	
Large DUT interface				x
DUT Scale interface	x	x	x	

- All DUT interfaces feature the "infinite plane" concept, which means any DUT board size fits on any DUT interface.
- On the DUT board/docking level, the large and the DUT Scale interface are identical.

Docking

Identical locations	A-testhead	Compact test head ^[7]	Small test head ^[7]	Large test head
Docking cylinders at front/back	none	x	x	x
Docking cylinders at tester sides (small DUT interface)	none	x	x	
Docking cylinders at tester sides (large DUT scale interface)	none	x	x	x



Manipulator

Manipulator type	A-test head ^[7]	Compact test head ^[7]	Small test head ^[7]	Large test head
A-test head type	x			
Compact test head type		x		
Small test head type			x	
Large test head type				x
Engineering cart	x	x		

Environmental

Operating	15°C to 30°C (59°F to 86°F)
Specification warranted temperature	20°C to 30°C (68°F to 86°F)
Maximum humidity at 30°C	< 70% R. H., non-condensing
Storage (without water)	-40°C to +70°C

Power requirements

This topic contains data about the various power requirements.

Maximum power consumption

Test head type	A-test head ^[8]	Compact test head ^[8]	Small test head ^[8]	Large test head
Maximum power	7 kVA	15 kVA	32 kVA	60 kVA

Need for auto-transformer

Test head type	A-, compact, small test head	Large test head	Condition
200V ± 10%	No	No	If # of Pin Scale cards ≤ 48 or If # of Smart Scale cards ^[9] ≤ 32
		Yes	If # of Pin Scale cards > 48 or If # of Smart Scale cards ^[9] > 32
208V ± 10%	No	No	If # of Pin Scale cards ≤ 48 or If # of Smart Scale cards ^[9] ≤ 32
		Yes	If # of Pin Scale cards > 48 or If # of Smart Scale cards ^[9] > 32
400V ± 10%	No	No	Independent of # of digital cards
440V ± 10%	Yes	Yes	Independent of # of digital cards
480V ± 10%	Yes	Yes	Independent of # of digital cards

^[8] Applies to test heads with the small DUT interface and with the DUT Scale interface.

^[9] Applies to Pin Scale 1600 and Pin Scale 9G

Cooling requirements

Cooling unit E2760FU Small liquid-liquid cooling unit

Cooling capability	6.6 kW @ 30°C regulated secondary water temperature
Cooling medium	De-ionized water
Primary chilled water temperature range	5°C to 20°C (41°F to 68°F)

- For use with one A-test head.

Cooling unit E2760FAL Liquid-liquid cooling unit for compact and small test heads

Cooling capability	28 kW @ 30°C regulated secondary water temperature
Cooling medium	De-ionized water
Primary chilled water temperature range	5°C to 20°C (41°F to 68°F)

- For use with one small / compact test head.

Cooling unit E2760FC Large liquid-liquid cooling unit mainframe

Cooling capability	56 kW @ 30°C regulated secondary water temperature
Cooling medium	De-ionized water
Primary chilled water temperature range	5°C to 20°C (41°F to 68°F)

- Recommended program value of secondary water temperature: 30°C.
- Large test head distribution unit (E2760FD) for use with one large test head.
- Quad-density distribution and interface unit for four cooler hose pairs (E2760FQ) for use with up to four compact test heads with independent shut down of test heads for repair or maintenance.

Weights and dimensions

Test system weight with manipulator and support rack (fully loaded)

A-test head	< 600 kg (no support rack needed)
Compact test head	< 1100 kg (no support rack needed)
Small test head	< 1500 kg
Large test head	< 2200 kg

Test head dimensions (height x width x depth, incl. docking)

A - test head (small DUT IF)	0.61 m x 0.59 m x 0.72 m
A - test head (DUT Scale IF)	0.64 m x 0.73 m x 0.72 m
Compact test head (small DUT IF)	0.61 m x 0.68 m x 0.85 m
Compact test head (DUT Scale IF)	0.64 m x 0.94 m x 0.85 m
Small test head (small DUT IF)	0.61 m x 0.77 m x 0.91 m
Small test head (DUT Scale IF)	0.64 m x 0.94 m x 0.91 m
Large test head	0.65 m x 1.02 m x 1.158 m

A - test head with engineering cart, incl. DUT interface, no hard docking

Dimensions	0.93 m (h) x 0.90 m (w) x 1.00 m (d)
Net weight	< 200 kg

Compact test head with engineering cart, incl. DUT interface, with & without hard docking

Dimensions (small DUT IF)	1.04 m (h) x 0.60 m (w) x 1.01 m (d)
Dimensions (DUT Scale IF)	1.04 m (h) x 0.73 m (w) x 1.01 m (d)
Net weight	< 310 kg

Cooling unit E2760FAL

Dimensions (height x width x depth)	1.14 m (h) x 0.65 m (w) x 1.05 m (d)
Net weight	175 kg

Cooling unit E2760FC incl. distribution unit

Dimensions (height x width x depth)	1.14 m (h) x 0.65 m (w) x 1.05 m (d)
Net weight	300 kg

Cooling unit E2760FU excluding hoses

Dimensions (height x width x depth)	0.65 m (h) x 0.24 m (w) x 0.42 m (d)
Net weight	40 kg